



Computer Engineering Department

Course Name: Microprocessor Lab

Number: 10636392

Lab Report Grading Sheet

Instructor: Dr.Aladdin Masri	Experiment #: 02
Academic Year: 2023/2024	Experiment Name: Counter/Timer using 8253
Semester: Summer semester	

Students				
1-Leen Imad Batta		2- Masa Anani		
3-		4-		
Performed on: 8/7/2024		Submitted on:15/7/2024		
Report's Outcomes				
ILO __ =() %	ILO __ =() %	ILO __ =() %	ILO __ =() %	ILO __ =() %
Evaluation Criterion			Grade	Points
Abstract answers of the questions: “What did you do? How did you do it? What did you find?”			0.5	
Introduction and Theory Sufficient, clear and complete statement of objectives. In addition to Presents sufficiently the theoretical basis.			1.5	
Apparatus/ Procedure Apparatus sufficiently described to enable another experimenter to identify the equipment needed to conduct the experiment.Procedure sufficiently described.			2	
Experimental Results and Discussion (In-Lab Worksheet) Crisp explanation of experimental results. Comparison of theoretical predictions to experimental results, including discussion of accuracy and error analysis in some cases.			4	
Conclusions and Recommendations Conclusions summarize the major findings from the experimental results with adequate specificity. Recommendations appropriate in light of conclusions. Correct grammar.			1	
Appearance Title page is complete, page numbers applied, content is well organized, correct spelling, fonts are consistent, good visual appeal.			1	
Total			10	



• Objectives

- To know the basic principle of 8253 and microcomputer interface.

To master the operation mode and program theory of 8253 timing/Counter.

• Abstracts

Utilizing the different modes of the 8235 counter to generate two waves with different frequencies and properties (square wave and rate generator wave).

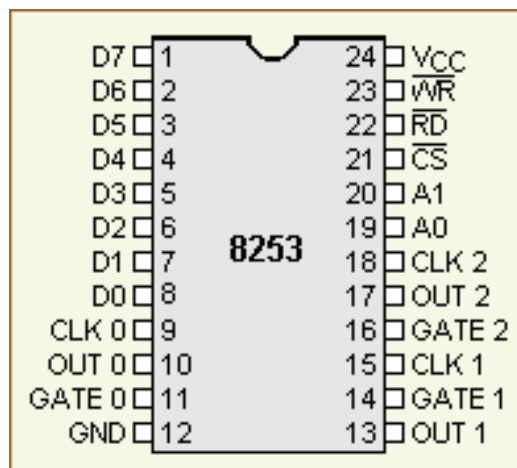
• Introduction

Intel 8253:

A Programmable Interval Timers (PTIs) designed for microprocessors to perform timing and counting functions using three 16-bit registers.

Hardware Specification of Intel 8253 - Programmable Interval Timer:

- It has 24 pins operating at +5 V dc.
- A maximum operating frequency of 2.6 MHz.
- It can handle inputs from DC to 10 MHz
- N-MOS technology.
- It consists of 3 Independent 16 bits down ↓ counters (0,1 and 2) that count either binary or BCD.
- Each counter has a clock input, gate input and output.
- Reads and writes of the same counter cannot be interleaved.
- Compatible with almost every microprocessor.





Programming the 8253:

The 8253 is programmed using a control word similar to the 8255. We specify the counter we want to use, its mode, and whether it counts BCD or binary. The A0, A1 inputs are used to select one of the counters or the control word (each one has a unique address).

Selects	Address
Counter 0	0040H
Counter 1	0041H
Counter 2	0042H
Control word register	0043H

Addresses table of the 8253

Control Word Format

D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀
SC1	SC0	RL1	RL0	M2	M1	M0	BCD

Definition of Control:

SC-Select Counter:

SC1	SC0	
0	0	Select counter 0
0	1	Select counter 1
1	0	Select counter 2
1	1	Illegal

M-Mode

M2	M1	M0	Mode
0	0	0	Mode 0
0	0	1	Mode 1
X	1	0	Mode 2
X	1	1	Mode 3
1	0	0	Mode 4
1	0	1	Mode 5

RL-Read/Load:

RL1	RL0	
0	0	Counter latching operation (see read/write procedure section)
1	0	Read/Load most significant byte only
0	1	Read/Load Least significant byte only
1	1	Read/Load Least significant byte first, then most significant byte

BCD:

BCD	
0	Binary counter 16-bits
1	Binary coded decimal (BCD) counter (4 decades)



In this exp, we are using the 8253 to output a square wave with 10KHz frequency. Since the 8253 outputs a 1MHz. We can divide it into 10KHz by using the frequency divider ($1\text{MHz}/10\text{KHz} = 100$)

● Procedures:

1. Connect the 8253 to the processor.
2. Write the associated code for each part.

Part 1:

In part one, we are using the 8253 counter 0 in mode 3 (square wave) to output a wave with a 10KHz frequency with an input frequency of 1MHz.

```
1 CODE SEGMENT
2 ASSUME CS:CODE
3 ORG 100H
4 START
5
6 mov dx,43h
7 mov al,16h
8 out dx,al
9 lm
10 mov dx,40h
11 mov al,100
12 out dx,al
13 jmp lm
14 END START
15 CODE ENDS
```

Part 2

In part 2, we are using the 8253 counter 0 in mode 2 (rate generator) to output a rate with a period of $100\ \mu\text{s}$ ($f=1/T = 1/100\ \mu\text{s} = 10\text{KHz}$).



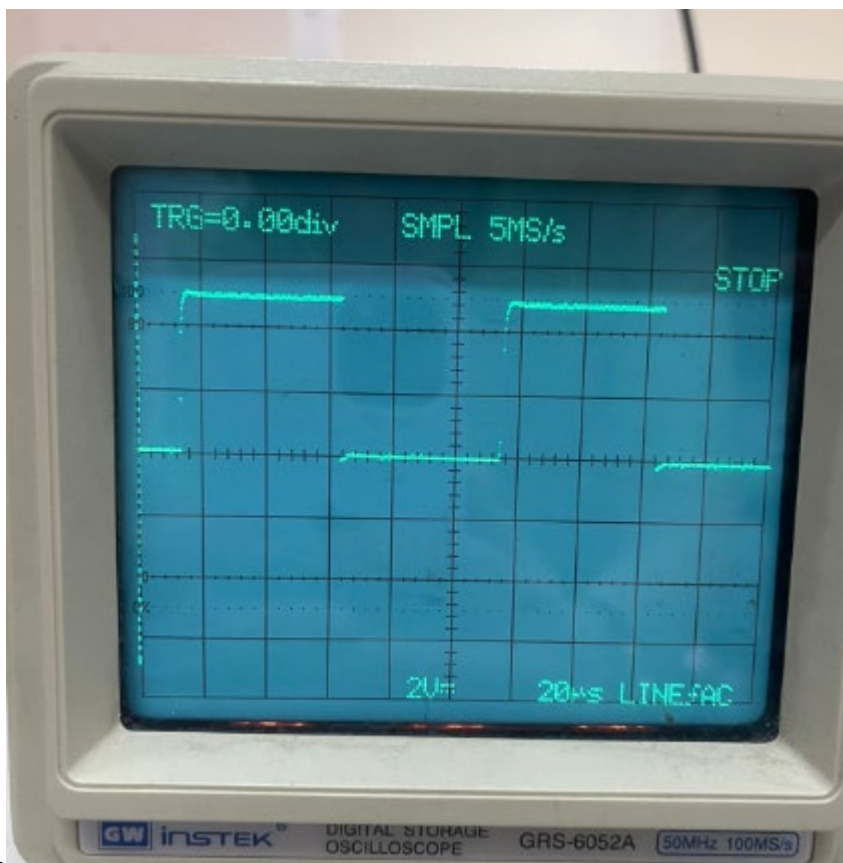
```
C:\Users\MPLab\Desktop\MExp2.asm
1 CODE SEGMENT
2 ASSUME CS:CODE
3 ORG 100H
4 START
5
6 mov dx,43h
7 mov al,14h
8 out dx,al
9 ln
10 mov dx,40h
11 mov al,100
12 out dx,al
13 jmp ln
14 END START
15 CODE ENDS
```

3-The 8253 output was connected to an oscilloscope for measurements and observation.

● Results:

Part1:

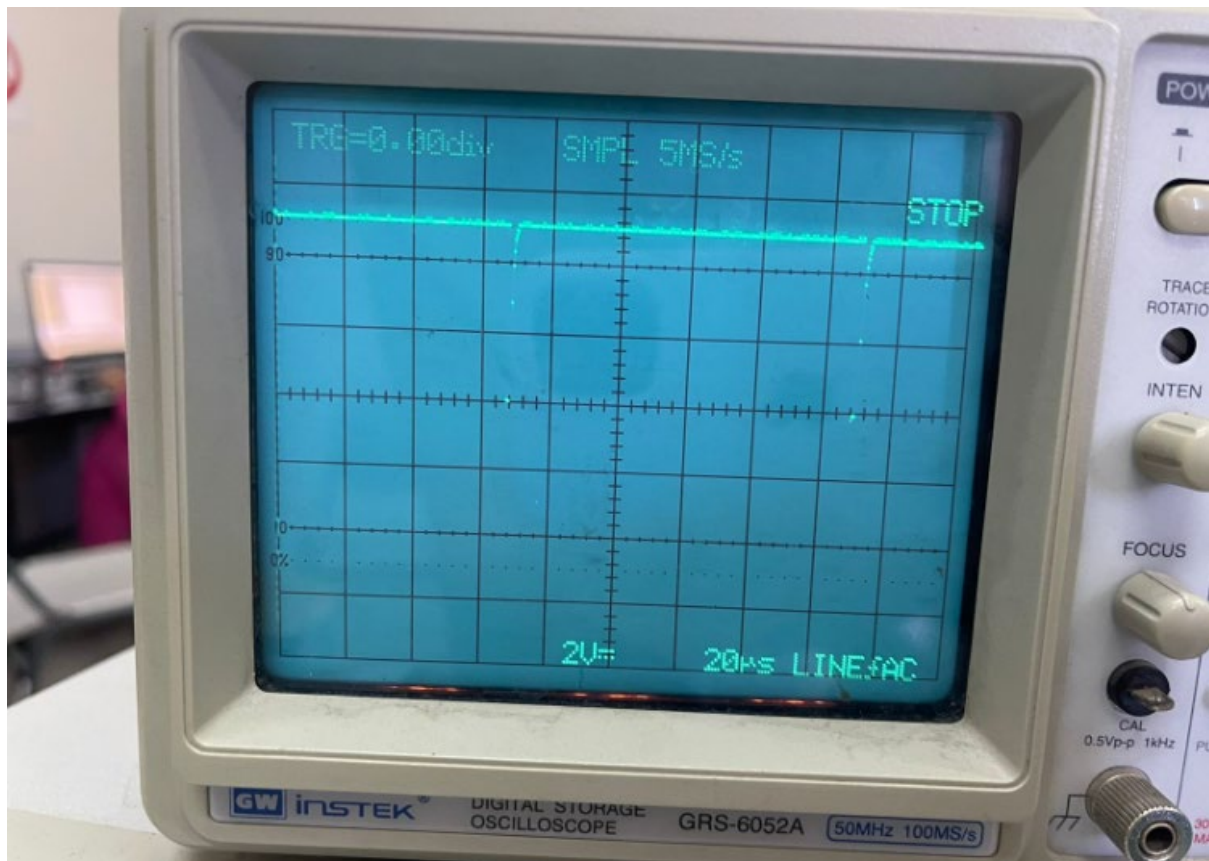
The output was a square wave with a 10KHz frequency.





Part 2:

The output was a 100 μ s rate.



• Conclusion:

We got to understand the operations of the 8253 counter and learned how to program and use it in different modes in order to generate different waves. The 8253 is similar to the 8255 IC in terms of programming since they both use a command word register.

• References

- Microprocessors Lab (10636392) manual
- Dr. Aladdin Masri lectures